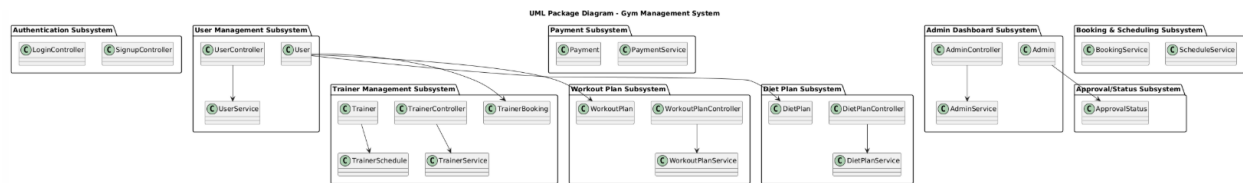


SYSTEM ARCHITECTURE

GYM MANAGEMENT SYSTEM

1) UML PACKAGE DIAGRAM



Subsystem Overviews:

- **Authentication Subsystem:** Handles user and admin login/signup processes, ensuring secure access to the system.
- **User Management Subsystem:** Manages user profiles, dashboards, and interactions with other modules like workout, diet, and trainer booking.
- **Trainer Management Subsystem:** Facilitates trainer details, availability schedules, and trainer-user bookings.
- **Workout and Diet Plan Subsystems:** Provide users with customized fitness routines and diet plans managed by trainers/admins.
- **Payment Subsystem:** Maintains records of membership payments and renewals.
- **Admin Dashboard Subsystem:** Allows admins to oversee system activity, approve users/trainers, and manage data.
- **Booking and Scheduling Subsystem:** Coordinates appointment slots and trainer-user schedules.
- **Approval/Status Subsystem:** Tracks and updates the approval status of users and trainers within the system.

2) ARCHITECTURAL STYLES

1. Layered Architecture

- Divides the system into presentation, business logic, and data access layers.
- Each layer communicates only with the layer directly below it.
 - **Controller Layer:** Handles incoming web requests (`UserController`, `AdminController`, etc.)
 - **Service Layer:** Processes business logic (`UserService`, `TrainerService`, etc.)

- **Repository Layer:** Interacts with the database (UserRepository, TrainerRepository).

2. Model-View-Controller (MVC)

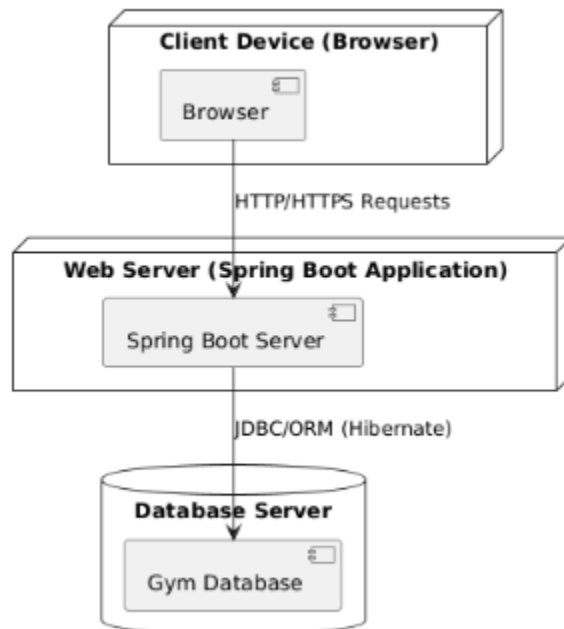
- **Model:** Java classes like User, Trainer, WorkoutPlan, representing database entities.
- **View:** HTML templates (user-dashboard.html, admin-dashboard.html) served to users.
- **Controller:** Java classes that receive HTTP requests and map them to services and views (UserController, TrainerController).
- This style clearly separates concerns between UI, backend logic, and data.

3. Repository Pattern

- Repositories abstract database interactions.
- Promotes cleaner code, testability, and easier future changes to data sources.

3) DEPLOYMENT DIAGRAM

Deployment Diagram - Gym Management System (Client Deployment)



4) COMPONENT DIAGRAM

